Beam–Beam Fields and Passive Pair Tracking for a Gamma–Gamma Collider

Current OPALX Model and Validation Status

A. Adelmann

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OPALX branch `271-implement-interation-point-element`, commit `df7cd6cba`

Abstract

Low-energy electron–positron pairs created near the interaction point of a gamma–gamma collider propagate through the collective electromagnetic fields of the two primary electron bunches. This note describes the current OPALX model for that interaction, the CAIN input data, an independent manufactured solution, and the present validation evidence. The current implementation supports exact timed witness emission, passive field sampling, a mirrored primary source, single- and multi-GPU execution, and an optimized one-solve field path. A 12-particle benchmark agrees with the like-for-like three-sigma-truncated manufactured trajectory to 0.6004 % in the transverse coordinate and 0.0971 % longitudinally. The remaining work is dominated by the multiscale production mesh: a physical 15 cm aperture must coexist with a primary transverse rms size of only 1.944 µm. A symmetric interaction window beginning and ending at a primary-centroid distance $d = 10\sigma_z$ from the interaction point is proposed as the first cutoff-convergence hypothesis.

1 Introduction and Problem Formulation

The interaction region contains two counter-propagating high-energy electron bunches, referred to here as the *primary beams*. Photons generated from the primary beams collide close to the interaction point (IP) and produce low-energy electron–positron pairs. These particles are referred to as *witnesses*: they sample the primary collective field, but their number is too small for their charge density or mutual interaction to be relevant in the present model. They therefore do not deposit charge and do not back-react on the primaries.

The immediate objective is not pair generation inside OPALX. Instead, CAIN phase-space records are imported with an individual birth time for every electron and positron. OPALX must then transport each witness from its exact birth event through the time-dependent fields of the two primary bunches. The production target contains 1,297 electrons and 1,297 positrons.

Figure **??** summarizes the model geometry. The IP is the midpoint of the placed `BEAMBEAM` element. In the current production geometry the element begins at $s = 1$ cm, is 32 cm long, and therefore places the IP at $s = 17$ cm. The second primary is represented by mirror symmetry about the IP, not by a second independently tracked particle container.

The reference primary parameters are collected in Table **??**. The primary support used for controlled manufactured comparisons is truncated independently in each coordinate at three rms sizes and renormalized.

For a witness of charge q and mass m_e, OPALX uses normalized momentum $\boldsymbol{P} = \gamma\boldsymbol{\beta}$. The

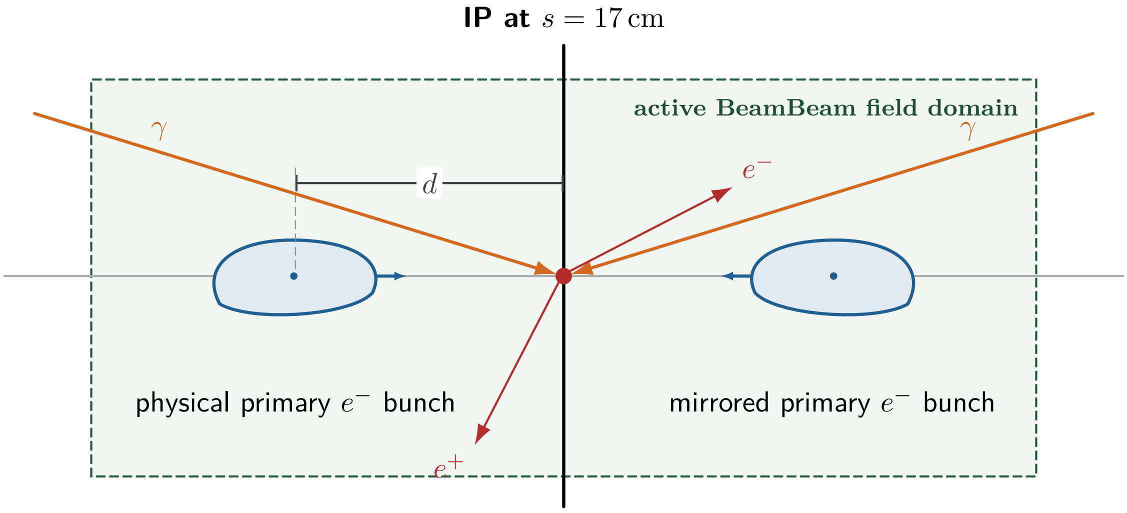

Figure 1: Beam–beam geometry. The primary and mirrored-primary centroids approach the IP from opposite sides. The distance d is defined from one primary centroid to the IP; the centroid-to-centroid separation is $2d$. Witness particles are released at their individual birth times and sample the primary fields without contributing to the field solve.

Table 1: Reference primary and production interaction-region parameters.

Quantity	Symbol or input	Value
Primary kinetic energy	K_0	$245\,\mathrm{MeV}$
Electrons per primary bunch	N_e	1.25×10^{10}
Transverse rms size	$\sigma_x = \sigma_y$	$1.944\,325\,075\,701\,\mu\mathrm{m}$
Longitudinal rms size	σ_z	$0.6\,\mathrm{mm}$
Controlled source support		$\lvert x_i \rvert \leq 3\sigma_i$
Production element length	L_{BB}	$32\,\mathrm{cm}$
Production aperture radius	R_{ap}	$15\,\mathrm{cm}$
Production witnesses		$1297\ e^- + 1297\ e^+$

continuous equations represented by the particle pusher are

$$\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} = c\frac{\boldsymbol{P}}{\sqrt{1+\boldsymbol{P}^2}}, \tag{1}$$

$$\frac{\mathrm{d}\boldsymbol{P}}{\mathrm{d}t} = \frac{q}{m_e c}\left(\boldsymbol{E} + c\frac{\boldsymbol{P}}{\sqrt{1+\boldsymbol{P}^2}} \times \boldsymbol{B}\right). \tag{2}$$

The fields are the superposition of the physical and mirrored primary contributions. Witness fields are excluded.

2 CAIN Results

CAIN supplies two datasets with complementary roles. The file `fort98.txt` contains the production population of 1,297 electrons and 1,297 positrons. The artificial Track-12 dataset contains six electron–positron pairs and densely sampled trajectories used for exact-timed code-to-code comparison.

In the converted production data, paired electrons and positrons have identical creation coordinates and creation times. Their momenta differ according to the pair kinematics. The creation times span −4.137 532–3.237 053 ps, have a standard deviation of 1.037 166 ps, and are equal within each electron–positron pair. The mean kinetic energies are 313.60 keV for both species. The momentum directions are broad: approximately half of each species travels with positive p_z and half with negative p_z.

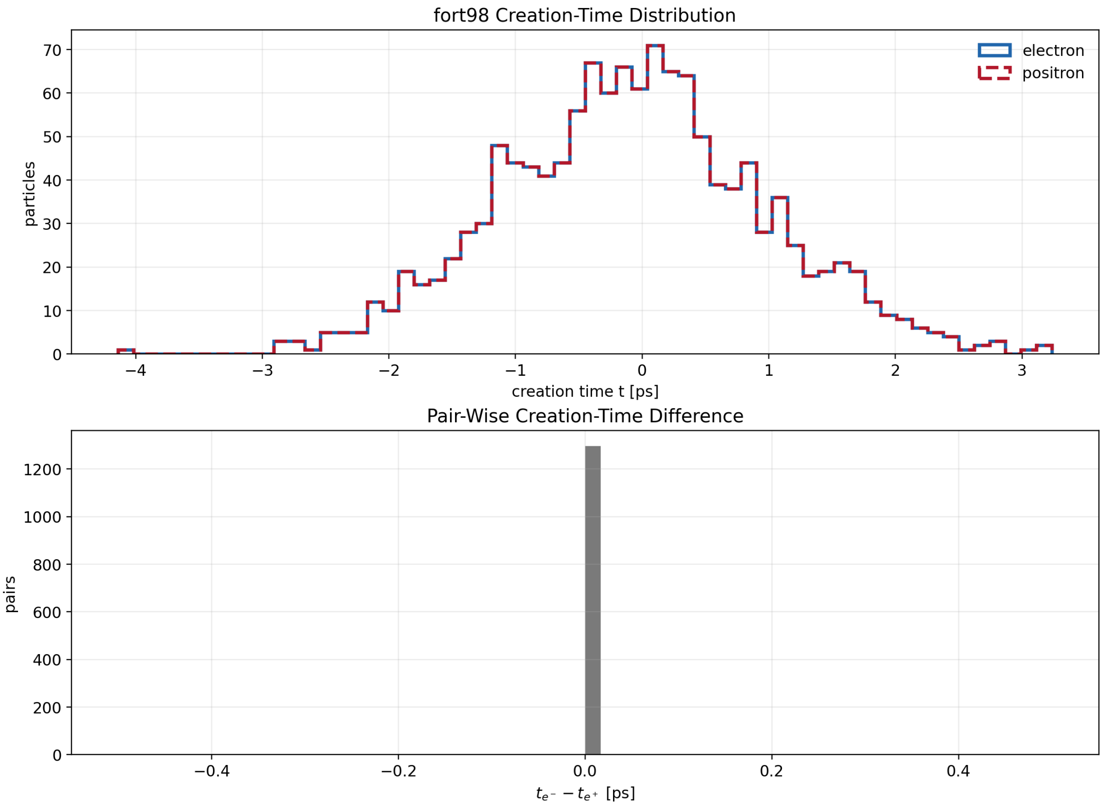

Figure 2: CAIN production-pair creation-time distribution. The upper panel shows the identical electron and positron time spectra; the lower panel verifies zero pair-wise time difference for all 1,297 pairs.

The Track-12 births occur at

$$ct = (-0.9000, -0.5994, -0.3006, 0, 0.3006, 0.5994)\ \mathrm{mm}. \tag{3}$$

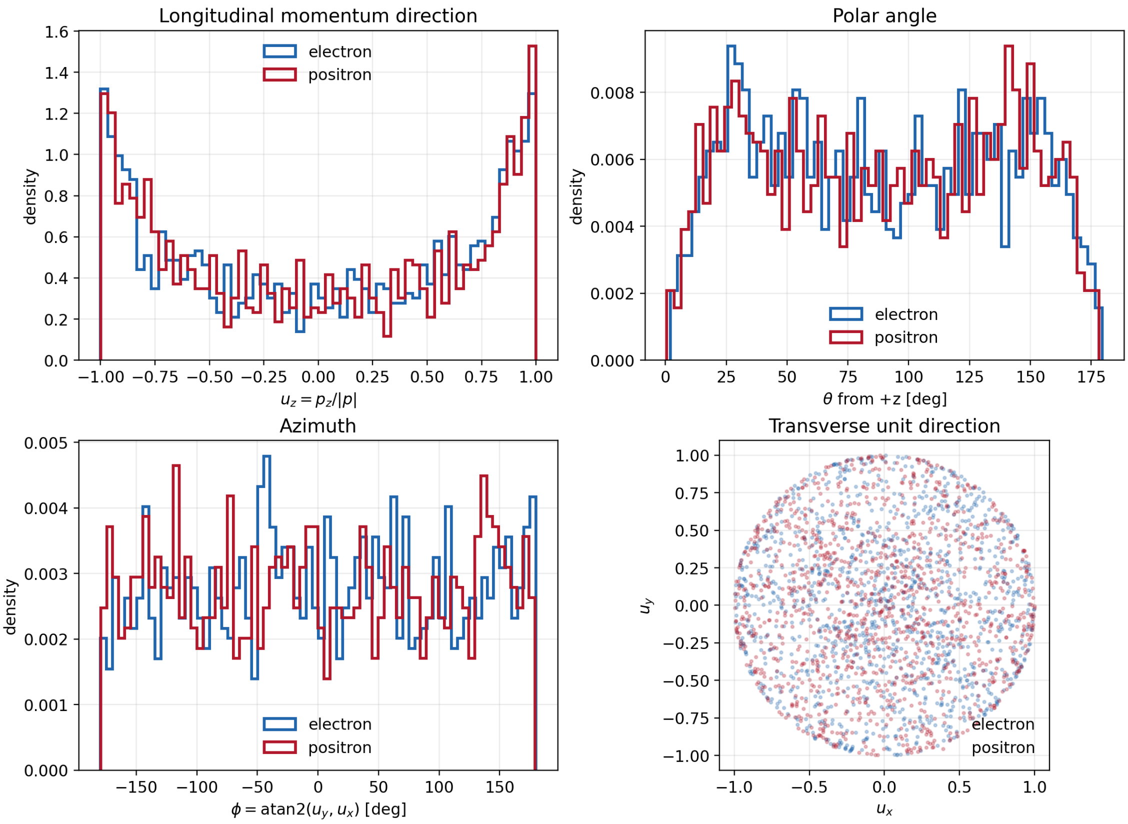

Figure 3: Direction distributions of the CAIN production witnesses. The broad angular support is important for the field-mesh coverage and aperture studies: the witness population is not a narrow paraxial beam.

CAIN samples every

$$\Delta t = \frac{1.8\,\mu\mathrm{m}}{c} = 6.004\,153\,713\,566\,736\,\mathrm{fs} \tag{4}$$

through $ct = 1.8\,\mathrm{mm}$. OPALX uses the same clock, so all 13,012 trajectory samples can be compared without temporal interpolation.

CAIN and the current OPALX manufactured model do not use identical collective field descriptions. CAIN is therefore a physics comparison, not the numerical oracle for the OPALX Poisson solve. Agreement with CAIN and agreement with the like-for-like manufactured solution must be reported separately.

3 Manufactured Solution

The independent manufactured solution represents two unchanging, uniformly moving anisotropic Gaussian primary bunches. Their centroids move in opposite longitudinal directions, and the IP is at the origin. For a source moving with velocity $\boldsymbol{v}_s = c\boldsymbol{\beta}_s$, the laboratory-frame event is transformed to the source rest frame. In the present straight geometry,

$$x' = x - x_c(t), \qquad y' = y - y_c(t), \qquad z' = \gamma_s[z - z_c(t)]. \tag{5}$$

The rest-frame rms sizes are

$$\sigma'_x = \sigma_x, \qquad \sigma'_y = \sigma_y, \qquad \sigma'_z = \gamma_s \sigma_z. \tag{6}$$

For an untruncated triaxial Gaussian, the rest-frame electric field is evaluated with the ellipsoidal integral

$$E'_i(\boldsymbol{r}') = \frac{Q x'_i}{4\pi\epsilon_0\sqrt{2\pi}} \int_0^\infty \frac{\exp\left[-\frac{1}{2}\sum_j \frac{x'^2_j}{\sigma'^2_j + u}\right]}{(\sigma'^2_i + u)\sqrt{(\sigma'^2_x + u)(\sigma'^2_y + u)(\sigma'^2_z + u)}}\,\mathrm{d}u. \tag{7}$$

The independent evaluator uses logarithmic quadrature in u, which resolves the large rest-frame aspect ratio. With $\boldsymbol{B}' = 0$, the laboratory fields are obtained from the uniform-motion Lorentz transformation. For a boost along z,

$$E_x = \gamma_s E'_x, \qquad E_y = \gamma_s E'_y, \qquad E_z = E'_z, \qquad \boldsymbol{B} = \frac{1}{c^2}\boldsymbol{v}_s \times \boldsymbol{E}. \tag{8}$$

Like-for-like OPALX comparisons use a component-wise three-sigma-truncated and renormalized density. This removes finite-support mismatch while retaining an independent field evaluator. At a three-σ_z centroid separation, a 20,301-probe comparison gave the results in Table **??**.

Table 2: Three-σ_z physical-plus-mirrored manufactured-field comparison.

Quantity	Value
OPALX maximum $\lvert\boldsymbol{E}\rvert$	5.545751×10^9 V/m
Manufactured maximum $\lvert\boldsymbol{E}\rvert$	5.553866×10^9 V/m
Relative L_2 error, $\boldsymbol{E}$	1.1098 %
Median field-magnitude ratio	0.997923
Median direction cosine	0.999970
Relative L_2 error, $\boldsymbol{B}$	1.1098 %

At exact centroid overlap, the identical sources have equal electric fields and opposite magnetic fields. The manufactured solution therefore predicts exact magnetic cancellation. The corresponding focused C++ regression verifies doubled deposited charge, restored physical coordinates, doubled electric field for a mirror-symmetric source, and cancelled magnetic field on one and two MPI ranks.

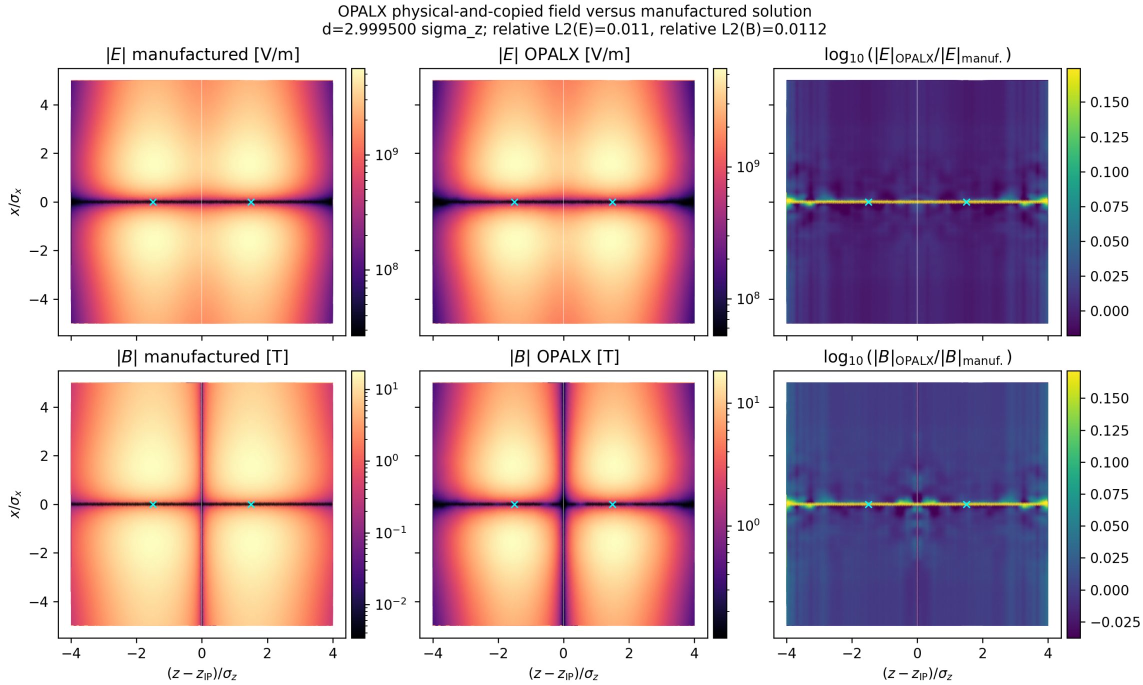

Figure 4: Representative three-σ_z comparison of the OPALX physical-plus-mirrored field with the independent anisotropic Gaussian manufactured solution.

4 Current OPALX Model

4.1 Software architecture

The status described here corresponds to branch `271-implement-interation-point-element` at commit `df7cd6cba`. The seven-step element redesign is complete. `BeamBeam` now follows the same ownership model as ordinary OPALX elements: the placed element contains input and placement configuration, while mutable per-run state belongs to `BeamBeamInteraction`. A generic `ElementInteractionManager` owns the interaction, and `ParallelTracker` dispatches generic `SelfField`, `AfterEmission`, and `Diagnostics` phases. The tracker no longer contains a BeamBeam-specific branch.

The IP is unconditionally the midpoint of the placed element. The former `IP_S` override has been removed. The current one-bin, no-cathode path uses only the persistent fields $\boldsymbol{E}_m$ and $\boldsymbol{B}_m$:

$$
\begin{aligned}
\text{primary particles} &\longrightarrow \rho \longrightarrow \text{one open-boundary Poisson solve} \longrightarrow (\boldsymbol{E}_p, \boldsymbol{B}_p) \\
&\longrightarrow \text{reflection about the IP} \longrightarrow (\boldsymbol{E}_v, \boldsymbol{B}_v) \\
&\longrightarrow (\boldsymbol{E}_m, \boldsymbol{B}_m) = (\boldsymbol{E}_p + \boldsymbol{E}_v, \boldsymbol{B}_p + \boldsymbol{B}_v).
\end{aligned}
$$

Reflection and accumulation execute in the active Kokkos memory space. The full fields are not staged through host memory. This path reduced ten-step solver calls from 20 to 11, where the remaining extra solve is warm-up, and reduced measured BeamBeam self-field time by 45.2 %.

4.2 Particle and emission contract

The three-container contract is

Container	Particles	Field role
0	primary electrons	deposits charge and normally receives the field
1	CAIN electrons	passive gather only
2	CAIN positrons	passive gather only

The `BBRIGID=TRUE` option suppresses the BeamBeam collective kick only on container 0. It does not suppress deposition, witness gathering, ordinary drift, or external fields. It is used for controlled comparison with the rigid manufactured source; production defaults to `BBRIGID=FALSE`.

Timed emitted files store

`x y z px py pz birth_time`.

The momentum components are $p/(m_e c)$ and the positions are relative to the emission-source origin. A witness born inside a step is integrated only over the fractional interval

$$\Delta t_{\text{particle}} = t_{\text{step end}} - t_{\text{birth}}. \tag{9}$$

Before gathering, each witness position is transformed from its independent container reference frame to the source-field frame. The gather then performs an MPI owner-rank reduction and transforms the sampled fields back.

4.3 Current fixed mesh and its limitation

During BeamBeam activity the field mesh is fixed to

$$x \in [-a_x, a_x], \qquad y \in [-a_y, a_y], \qquad z \in [z_{\text{IP}} - L/2, z_{\text{IP}} + L/2]. \tag{10}$$

For the production geometry this is a rectangular box of 30 cm × 30 cm × 32 cm enclosing the cylindrical aperture. The implementation currently checks only the transverse extent of container 0. It does not prove containment of the mirrored source or witness containers, and it does not check longitudinal containment.

More importantly, geometric coverage is not numerical resolution. The current large-cylinder debug deck uses 16^3 cells, corresponding to cell sizes of approximately 20 mm, 20 mm, and 21.33 mm. Even a $4096 \times 256 \times 128$ mesh over the same box would have spacings of approximately 73.3 µm, 1.176 mm, and 2.52 mm; it would not resolve the 1.944 µm transverse primary or the 0.6 mm longitudinal primary. The physical aperture and numerical field domain must therefore be separated.

5 Current OPALX Results

The current validation hierarchy is summarized in Table **??**. The complete local build and focused tests pass; Kokkos version 5.2.0 is configured. MPI-sensitive paths have been exercised on one, two, and four ranks.

Table 3: Selected validation and performance evidence.

Test	Particles	Mesh	Result	Wall time
Persistent manufactured CTest	100,000 source; 5,151 probes	$64 \times 64 \times 128$	Relative $L_2(\boldsymbol{E}, \boldsymbol{B}) = (2.10, 2.20)\%$	~ 5 s
Fixed-source rank scan	400,000 source; 5,151 probes	$64 \times 64 \times 128$	Four-rank fields agree with one rank to 2.4×10^{-16}	4.94–10.50 s
Witness-gather rank regression	100,000 source; $8e^- + 8e^+$	$32 \times 32 \times 64$	Rank differences in field and kick below 1.5×10^{-15}	5.85–9.82 s
Fine Track-12, four A100s	400,000 source; $6e^- + 6e^+$	$4096 \times 256 \times 128$	1,501 steps; all 13,012 samples; normal completion	25 278.83 s

The transverse first-kick refinement keeps the field domain fixed at 2.4 mm by 0.24 mm and $N_z = 128$. Refinement changes the field magnitude while the direction cosine remains at least 0.999993. The observed trend approaches second order, but anisotropic refinement matters: increasing N_x while coarsening N_y does not guarantee a smaller error.

For the full $4096 \times 256 \times 128$, 1,501-step trajectory, the principal comparison is

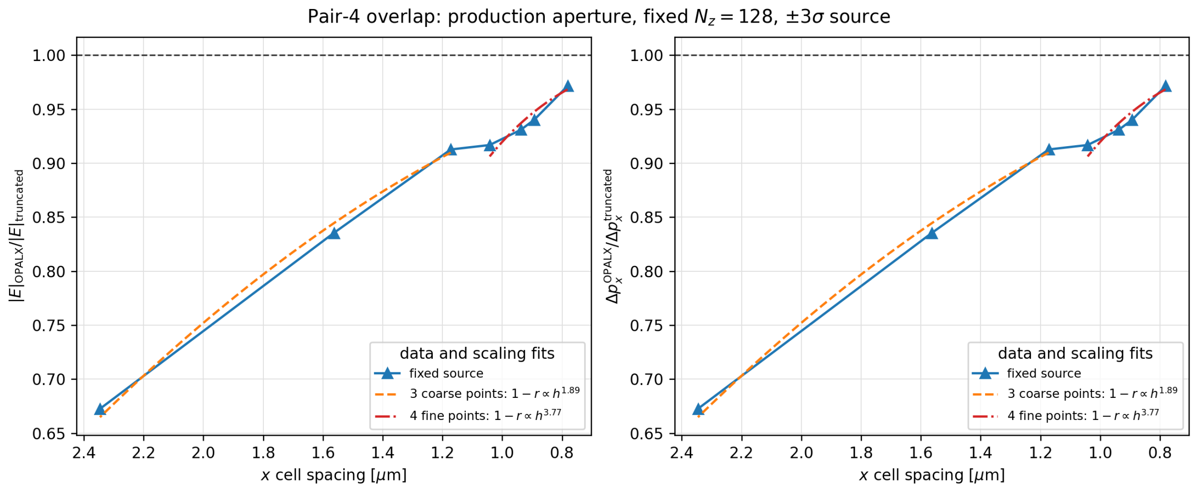

Figure 5: Pair-4 electric-field and first-kick convergence toward the three-sigma-truncated manufactured source. The first three coarse points and last four fine points are fitted separately.

Reference	x relative L_2	x RMSE	longitudinal relative L_2	longitudinal RMSE
Manufactured	0.6004 %	1.996 µm	0.0971 %	0.807 µm
CAIN	9.545 %	30.648 µm	2.9774 %	24.477 µm

The twelve first-x-kick ratios relative to the truncated manufactured model span 0.92947–0.97490, with mean 0.94860. Charge-conjugate OPALX kicks agree to a maximum relative residual of 7.57×10^{-10}. This supports the timed input, species assignment, charge sign, MPI gather, and fractional integration. The larger CAIN discrepancy is attributed to a difference in collective source model or convention and must not be tuned away in OPALX.

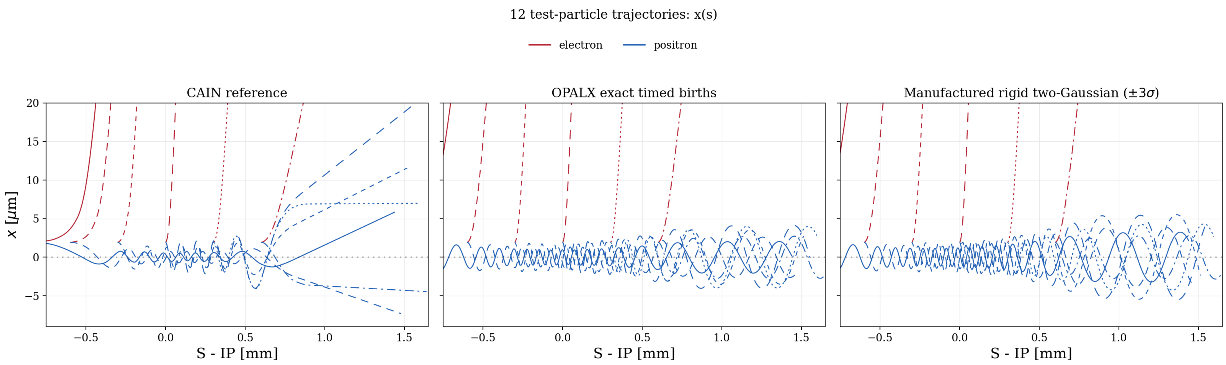

Figure 6: Latest four-A100 Track-12 comparison. CAIN is shown at left, OPALX with exact timed births in the centre, and the three-sigma-truncated rigid two-Gaussian manufactured solution at right.

The optimized two-field result changes the previous two-solve trajectory only at accumulated numerical levels: the maximum position changes are below 0.022 µm, and relative L_2 momentum differences are below 4×10^{-6}. The optimized full trajectory is 1.992 times faster than the otherwise identical two-solve run.

6 Next Steps in Refining the OPALX Model

6.1 Define a physical interaction cutoff

The first hypothesis is a symmetric interaction window defined by the distance of either primary centroid from the IP,

$$d = 10\sigma_z = 6\,\text{mm}. \tag{11}$$

With the 245 MeV primary, $\beta = 0.999997834$, so

$$\frac{d}{\beta c} = 20.0\,\text{ps}. \tag{12}$$

The mirrored interaction would begin approximately 20 ps before centroid overlap and end approximately 20 ps after it. At activation the centroid separation is $2d = 12\,\text{mm}$. A three-sigma-truncated source has its nearest edge seven rms lengths from the IP.

The intended timeline is therefore:

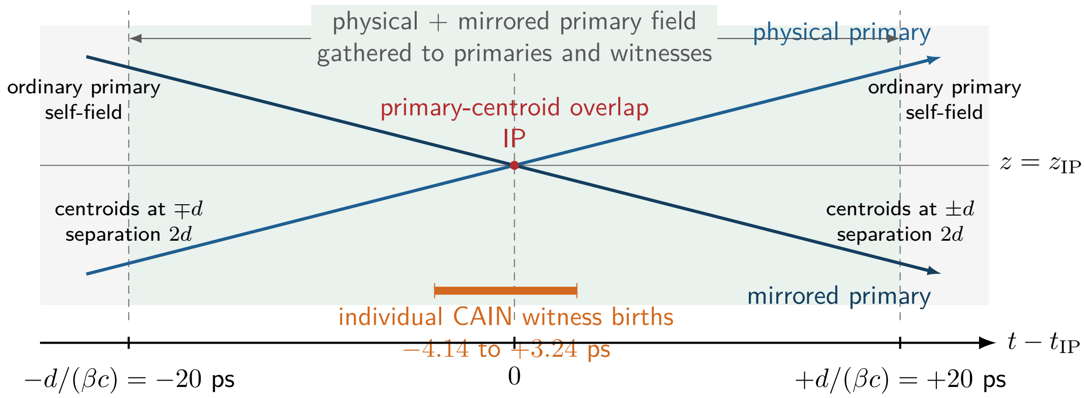

Figure 7: Proposed symmetric BeamBeam interaction timeline for $d = 10\sigma_z$. The combined physical-plus-mirrored field is enabled while either primary centroid is within distance d of the IP. Witnesses appear at their individual CAIN birth times and remain passive field gatherers. The drawing is a proposed cutoff model, not a statement of converged accuracy.

This is a cutoff hypothesis, not an accuracy result. Gaussian density at the IP is exponentially small at ten rms lengths, but the electromagnetic field is long-ranged. The integrated witness and primary impulses must be converged for $d/\sigma_z = 5, 7.5, 10, 15$, and 20. The comparison quantities should include final position and momentum, maximum transverse kick, primary moments at the IP, and aperture-loss count and location.

The implementation semantics also require refinement. Before the symmetric window the physical primary may continue its ordinary self-field evolution. Inside it, the physical and mirrored fields are combined and gathered to the witnesses. After it, the mirrored BeamBeam contribution and witness gather are disabled. This is not identical to the current `COPY_TIME`, which only enables the mirror, or `RETIRE_TIME`, which deletes the physical source. A geometric interaction-distance parameter is preferable to two absolute times tied to the upstream tracking origin.

6.2 Separate physical and numerical domains

For $d = 10\sigma_z$ and a three-sigma source, the longitudinal numerical domain must satisfy

$$z_{\text{half}} \gtrsim d + 3\sigma_z = 13\sigma_z = 7.8\,\text{mm}, \tag{13}$$

plus a mesh and interpolation margin. This suggests a numerical field length of approximately 16–18 mm, not the full 32 cm physical element.

The next code-level diagnostic should compute global minimum and maximum positions for all active containers in the source-field frame on every BeamBeam solve. It should report the distance to each field boundary in metres and cells and fail clearly when a source or gathered witness lies outside the supported interpolation region. The current primary-only transverse check is insufficient.

The transverse scale separation remains harder. A witness may move several millimetres during a 40 ps interaction window, whereas the primary transverse rms size is only 1.944 µm. A single uniform mesh that covers both scales and resolves the primary with several cells per rms size is expensive. Candidate refinements are a compact source mesh with a separate far-field evaluator, nested meshes, or a controlled analytic rigid Gaussian field outside the resolved source region. Each option must preserve the non-rigid production path and be validated against the manufactured solution.

6.3 Production-readiness sequence

Before interpreting a full 1,297-pair production run, the following sequence is required:

1. Align the timed CAIN file origin with the primary centroid-overlap clock and migrate the large-cylinder input from common-`TO FROMFILE` injection to the exact timed files.
2. Add source, mirror, and witness mesh-coverage diagnostics and a regression for out-of-domain gathering.
3. Decouple physical element length and aperture from the numerical field domain.
4. Converge the interaction cutoff d, timestep, transverse and longitudinal mesh, primary macroparticle count, and source truncation.
5. Repeat the selected production case on one and multiple A100s and verify decomposition independence of fields, trajectories, and losses.
6. Correct the known H5 electric-field unit metadata, which currently labels values stored in V/m as MV/m, after auditing downstream readers.

The first complete production run should therefore be treated as a workflow and coverage test. Physics conclusions about pair losses should be drawn only after the cutoff and discretization studies are complete.

A Reproducing the Data, Figures, and OPALX Runs

This appendix records the workflow used for the results in this note. Unless stated otherwise, commands are run from the root of the `opalx-beambeam` checkout with the Python environment `$HOME/.venv-h6`. Generated H5 files and plots are intentionally ignored by Git. A reproducible run must retain the Git revision, input hashes, mesh, timestep, particle count, MPI rank count, GPU identity, executable hash, `runtime_compute.txt`, nonempty `timing.dat`, and the scheduler log.

The results reported here correspond to the source state identified in Section 4.1. To reproduce a historical number exactly, check out that revision and do not reuse input or manifest files from a different mesh. For new studies, use the current branch tip and record the resulting revision in the run manifest.

A.1 Figure provenance

Table **??** maps every figure in this note to its source or generating program. Figure **??** is native TikZ in this LaTeX file and is regenerated by compiling the note.

Table 4: Programs and principal output files for the figures in this note.

Figure	Program or source	Output used by this note
??	`bb-note.tex`	inline TikZ, generated during LaTeX compilation
??	`analyze_pair_momenta.py`	`pair_momentum_time_histogram.png`
??	`analyze_pair_momenta.py`	`pair_momentum_directions.png`
??	`compare_opalx_fields.py`	`3sigma_physical_and_copied_field_comparison.png`
??	`scan_first_kick_fields.py`	`pair4_field_and_kick_convergence.png`
??	`compare_timed_track12.py`	`results/track12_cain_vs_opalx.png`
??	`bb-note.tex`	inline TikZ, generated during LaTeX compilation

A.2 CAIN pair data and Figures ??–??

The full CAIN pair table is `sandbox/track-e-p/fort98.txt`. Convert it to separate OPALX electron and positron files while retaining the individual creation times, then make the statistical tables and figures as follows:

```
cd sandbox/track-e-p
$HOME/.venv-h6/bin/python convert_fort98_to_fromfile.py \
  fort98.txt --add-time -o .
$HOME/.venv-h6/bin/python analyze_pair_momenta.py \
  --electron-file gamma_gamma_electrons-t.fromfile \
  --positron-file gamma_gamma_positrons-t.fromfile \
  --output-dir data --plot
cd ../..
```

The second command writes the two PNG files used in this note, CSV summaries, a representative pair-geometry plot, and a multipage PDF. Species pairing is by row order from the CAIN event table; the conversion should therefore be rerun whenever `fort98.txt` changes.

A.3 Independent manufactured solution

The independent field evaluator implements the Lorentz transformation of an anisotropic Gaussian at uniform velocity. It does not call the OPALX field solver. Generate the smooth field snapshots at centroid separations $d/\sigma_z = 3, 2, 1, 0$ and run its focused tests with

```
$HOME/.venv-h6/bin/python \
  sandbox/reduced-order-model/python/rigid_two_gaussian_fields.py
$HOME/.venv-h6/bin/python -m unittest \
  sandbox/reduced-order-model/python/test_rigid_two_gaussian_fields.py \
  sandbox/reduced-order-model/python/test_fixed_primary_fromfile.py
```

Within `sandbox/reduced-order-model`, the field tables, summary JSON, and overview plot are written below `outputs/fields`; the authoritative inputs are in `parameters.json`. Electric fields are in V/m, magnetic fields in tesla, positions in metres, and momenta in $m_e c$ units.

For the OPALX comparison in Figure **??**, first generate and run the matching OPALX probe cases locally, then compare the two-source field after mirroring:

```
$HOME/.venv-h6/bin/python \
  sandbox/reduced-order-model/python/make_opalx_field_cases.py --run
$HOME/.venv-h6/bin/python \
  sandbox/reduced-order-model/python/compare_opalx_fields.py \
  --cases 3sigma --step 1 \
  --source-model physical-and-copied \
  --output-dir \
    sandbox/reduced-order-model/outputs/comparison-two-source
```

The default generator uses 400,000 primary macroparticles, a $64 \times 64 \times 128$ field mesh, and a 101×201 passive-probe grid. The odd probe dimensions place a sample exactly at the IP. `Step#0` is the physical bunch alone; `Step#1` is the physical-plus-mirrored field. The OPALX deck must use `GREENSF=INTEGRATED`, `BBRIGID=TRUE`, and `EBDUMP=TRUE` for this comparison.

The manufactured Track-12 trajectories are recomputed by `compare_timed_track12.py`; they are not read from a stored trajectory. The integrator uses both anisotropic sources, component-wise three-sigma source truncation, exact CAIN birth times, and a maximum internal substep of 1 fs.

A.4 Building OPALX for Merlin6

The GPU calculations use the persistent Merlin6 checkout and its configured Release build. Login-node availability can be intermittent, so all expensive work is submitted through Slurm and all scripts and logs are stored inside the run directory. On Merlin6:

```
ssh merlin6
SRC=/psi/home/adelmann/opalx-dev/opalx-a100-5a354101f
cd "$SRC"
git status --short
git rev-parse HEAD
cmake --build build-a100 -j20
grep CMAKE_BUILD_TYPE build-a100/CMakeCache.txt
OPALX="$SRC/build-a100/src/opalx"
test -x "$OPALX"
```

The intended configuration is Release with CUDA and Ampere-80 support. If the existing build directory is unavailable, it must be configured with the Merlin site dependency prefixes before the build command above; those paths are installation-specific and should not be guessed. Record the CMake cache and `sha256sum $OPALX` in a fresh production campaign.

A.5 Manufactured-field A100 study on Merlin6

The self-submitting convergence driver prepares its input matrix, launches the single-GPU cases, then connected two- and four-rank MPI cases, and finally checks completion. It copies scheduler output into `RUN_ROOT/slurm`.

```
cd "$SRC"
RUN_ROOT=/psi/home/adelmann/opalx-dev/\
bb-manufactured-$(git rev-parse --short HEAD)-$(date -u +%Y%m%dT%H%M%SZ)
sandbox/reduced-order-model/merlin/submit_a100_convergence.sh \
  --run-root "$RUN_ROOT" --repo-root "$SRC" --opalx "$OPALX"
squeue -M gmerlin6
```

Connected multi-GPU runs use `mpiexec` with one rank per A100 and `--kokkos-map-device-id-by=mpi_rank`. Plain multi-task `srun` must not launch OPALX on this cluster because it creates independent Open MPI singletons. Independent one-rank cases may run as exclusive `srun` steps inside a GPU allocation, as implemented by the driver.

After completion, return to the local checkout and fetch only compact products before plotting:

```
$HOME/.venv-h6/bin/python \
  sandbox/reduced-order-model/python/fetch_and_plot_a100_convergence.py \
  --remote-host merlin6 --remote-run-root "$RUN_ROOT"
```

Because a local shell does not inherit the remote `RUN_ROOT`, replace it with the absolute study path printed by the submission script.

A.6 Exact-timed Track-12 run on four A100s

Prepare the deterministic 400,000-particle, $4096 \times 256 \times 128$ case on Merlin and stage only the required files into a new run-local production directory:

```
cd "$SRC"
PY=python3
TIMED=sandbox/track12particles/opalx/timed
$PY "$TIMED/prepare_timed_track12.py" \
  --primary-macroparticles 400000 \
  --nx 4096 --ny 256 --nz 128

RUN_ROOT=/psi/home/adelmann/opalx-dev/\
track12-4096x256x128-4rank-$(git rev-parse --short HEAD)-\
$(date -u +%Y%m%dT%H%M%SZ)
mkdir -p "$RUN_ROOT/production/results"
cp "$TIMED/track12_timed.in" "$RUN_ROOT/production/"
cp -R "$TIMED/input" "$RUN_ROOT/production/"
cp "$TIMED/results/preparation_manifest.json" \
  "$RUN_ROOT/production/results/"

"$TIMED/merlin/submit_track12_a100.sh" submit \
  --run-root "$RUN_ROOT" --opalx "$OPALX" \
  --mpi-ranks 4 --omp-threads 4 --production-time 10:00:00
squeue -M gmerlin6
```

The launcher first submits a three-step smoke test and submits the 1,501-step calculation with an `afterok` dependency. The actual OPALX command uses `mpiexec -n 4`, `--info 1`, and `--kokkos-map-device-id-by=mpi_rank`. The nonzero information level is required for a populated `timing.dat`. A `completed` marker is written only after OPALX exits successfully and `timing.dat` is nonempty. The exact submitted launcher and the `smoke_%j.out` and `production_%j.out` logs remain under `RUN_ROOT/slurm`.

Fetch the finished witnesses to the local machine and regenerate Figure ?? and all comparison tables with

```
DEST=sandbox/track12particles/opalx/timed/\
a100_4rank_400k_4096x256x128_current
$HOME/.venv-h6/bin/python \
  sandbox/track12particles/opalx/timed/fetch_and_compare_a100.py \
  --host merlin6 --remote-dir "$RUN_ROOT/production" \
  --destination "$DEST"
```

Again, substitute the absolute remote run path for `$RUN_ROOT` in the local shell. The comparison requires both witness H5 files, the exact input deck, emitted witness files, and the matching preparation manifest. It writes the trajectory PNG, pointwise CSV, first-kick CSV, particle summary, and JSON metrics below `DEST/results`.

A.7 Reproducing Figure ??

For already-fetched field-probe cases, regenerate the convergence CSV and plot without rerunning OPALX:

```
OUT=sandbox/track12particles/opalx/timed/\
a100_mesh_8da5b9e83_1rank
$HOME/.venv-h6/bin/python \
  sandbox/track12particles/opalx/timed/scan_first_kick_fields.py \
  --analyze-only --output-dir "$OUT" \
  --fixed-primary "$OUT/input/primary_fixed.fromfile"
```

For new A100 data, first prepare the desired named cases with the same driver, stage them under a fresh Merlin run root, and submit the tracked `submit_first_kick_field_probe_a100.sh` launcher. Fetch with `fetch_and_plot_first_kick_a100.py`. Do not combine curves produced from different fixed-primary hashes, field-domain sizes, longitudinal meshes, or source truncations without labeling those differences explicitly.

A.8 Recompiling this note

The final paper draft is compiled from the repository root with

```
cd sandbox/note
latexmk -pdf -interaction=nonstopmode -halt-on-error bb-note.tex
cd ../..
cp sandbox/note/bb-note.pdf output/pdf/bb-note.pdf
```

All referenced PNG files must exist before compilation. The PDF should then be rendered page-by-page and inspected for clipped listings, tables, figures, and equations.